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AUTO DISABLE PREFILLED SYRINGE WITH PLUNGER ROD SPRING

Abstract

A syringe may include a barrel with a needle. A plunger stopper may be disposed within the barrel, and displaceable along an axis of the barrel. A plunger rod may be provided with an elongated rod, and adapted to be inserted into the barrel. The plunger rod may be adapted to transmit force to the plunger stopper. A radial spring mechanism may be provided on the elongated rod, and may interact with an inner surface of the barrel. The radial spring mechanism may be located radially outward from a center portion of the elongated rod. The center portion of the elongated rod may extend through the radial spring mechanism. The plunger stopper may be adapted to receive force from the plunger rod. The plunger rod may not be connected to the plunger stopper, such that the plunger stopper cannot be pulled back by the plunger rod.

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Background/Summary

[0001] This application is a continuation of U.S. patent application Ser. No. 17/044,135 filed Sep. 30, 2020, which is a US National Phase Application of International Application No. PCT/EP2019/057103 filed Mar. 21, 2019, the entire contents of both of which are incorporated herein by reference. [0002] This application claims priority under 35 USC 119 (a)-(d) from SE Patent Application No. 1850388-8 filed on Apr. 6, 2018, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0003] The present invention relates to a prefilled syringe with a construction that prevents re-usage of the syringe and wherein the syringe may be manufactured at a cost similar to a conventional prefilled syringe.

2. Discussion of Related Art

[0004] The business area of prefilled syringes is undergoing strong expansion. There are several reasons for this, but one of the main reasons is that the prefilled syringes are “ready for use”. Therefore, the need for the users to fill the syringes and the handling of the drugs and syringes before use is minimized. This will accordingly reduce the risk of infections and assure the quantity and the quality of the drug contained within the syringe.

[0005] Prefilled syringes are usually made of glass with highest quality. The reason for this is that the drug contained within the syringe carries the highest cost. The syringes are constructed such that they have a minimal dead space, which will assure that almost all drug or medicine contained within the syringe will be released. However, these positive characteristics of the syringes will make it tempting for users to reuse them, as conventional syringes, by refilling the syringes with drugs again. Misusage of syringes is a common issue; over 2 million users each year are infected with severe blood transmittable diseases, such as HIV, Hepatitis B and C. By current constructions and designs of prefilled syringes, this problem is not completely solved, and the attempts to solve the problem are done at a very high cost.

[0006] Hence, there is a need to provide a syringe which addresses the disadvantages and shortcomings described above and to provide a non-reusable syringe.

SUMMARY

[0007] Accordingly, the present invention preferably seeks to mitigate, alleviate or eliminate one or more of the above-identified deficiencies in the art and disadvantages singly or in any combination and solves at least the above mentioned problems by proposing a solution according to the appended independent claims. Advantageous embodiments are defined in the appended dependent claims.

[0008] In a first of its aspects, this disclosure therefore presents a plunger rod for use in connection with an injection syringe barrel. The plunger rod comprises an elongated rod. The elongated rod comprises a proximal end adapted to receive an applied force along an axis. It further comprises a

distal end adapted to transmit the applied force along the axis. The plunger rod is adapted to be inserted into a barrel of a syringe extending along the axis. The plunger rod is further adapted to transmit force along the axis to a plunger stopper disposed within the barrel. The plunger rod further comprises a radial spring mechanism, adapted to interact with an inner surface of the barrel. [0009] One advantage of this is that the plunger rod may be manufactured separately from the syringe. Another advantage of this is that the plunger rod may be recycled or more efficiently disposed of, since it is not damaged when removed from the barrel. Another advantage of this is that since the plunger rod is not connected to the plunger stopper, the plunger stopper cannot be pulled back. An advantage of the radial spring mechanism is that the plunger rod will be stabilized and directed by the inner surface of the barrel.

[0010] In some embodiments, the plunger rod is made of a thermoplastic polymer having a Young's modulus within the interval 500 MPa to 2000 MPa.

[0011] In some embodiments, the rod of the plunger rod has a cross-shaped cross-section comprising four supporting ribs extending radially from the center of the rod. At least one supporting rib comprises the spring mechanism.

[0012] One advantage of this is that the plunger rod will have a smaller friction engaging surface and requires less material to manufacture.

[0013] In some embodiments, the four supporting ribs comprise interaction knobs.

[0014] One advantage of this is that it may be easier to optimize the applied spring force.

[0015] In some embodiments, the four supporting ribs comprise interaction discs.

[0016] One advantage of this is that it may be easier to optimize the applied spring force.

[0017] In some embodiments, the distal end of the plunger rod comprises a spring portion adapted to being displaced radially from their relaxed position once the plunger rod is inserted into said barrel of said syringe.

[0018] One advantage of this is that the spring portion allows for simple adjustment of the friction force of the supporting rib against the inner surface of the barrel.

[0019] In some embodiments, at least one of said four supporting ribs comprise spring portions adapted to being displaced radially from their relaxed position once the plunger rod is inserted into said barrel of said syringe.

[0020] One advantage of this is that the spring portion allows for simple adjustment of the friction force of the supporting rib against the inner surface of the barrel.

[0021] In some embodiments, the spring portion comprises an axial slit.

[0022] One advantage of this is that it may be easier to optimize the applied spring force.

[0023] In some embodiments, the axial slit is arranged in the middle of the rod.

[0024] One advantage of this is that it may be easier to optimize the applied spring force.

[0025] In some embodiments, the axial slit is contained within said at least one supporting rib.

[0026] One advantage of this is that there is no risk for the slit to interact with a proximal end of the barrel upon insertion of the plunger rod into the barrel.

[0027] In some embodiments, the spring portion comprises an interaction spoke.

[0028] One advantage of this is that there is a smaller risk for the interaction spoke to interact with a proximal end of the barrel upon insertion of the plunger rod into the barrel.

[0029] In some embodiments, the spring portion comprises a set of reeds.

[0030] One advantage of this is that the reeds may more easily conform with the inner surface of the barrel. There is also a smaller risk for the reeds to interact with a proximal end of the barrel upon insertion of the plunger rod into the barrel.

[0031] In a second of its aspects, this disclosure presents a syringe. The syringe comprises a barrel with a needle arranged at a distal end of the barrel. The barrel is adapted to contain a drug. The barrel has a tubular wall extending proximally from the distal end, whereby the barrel extends along an axis. The syringe further comprises a plunger stopper disposed within the barrel. The plunger stopper is adapted to be in fluid-tight engagement with the tubular wall of the barrel. The

plunger stopper is displaceable along the axis within the barrel. The barrel comprises an opening arranged at the proximal end of the barrel. The opening is adapted to receive a plunger rod according to a first aspect of this disclosure extending along the axis. The plunger stopper is adapted to receive force along the axis from said plunger rod.

[0032] In some embodiments, the syringe contains the drug.

[0033] One advantage of this is that pre-filled syringes reduce risk of spillage, overdosing and contamination.

[0034] In some embodiments, the plunger stopper is made of an elastomer having a Young's modulus within the interval 25 MPa to 300 MPa.

[0035] In some embodiments, the plunger stopper is adapted to only be displaceable along the axis towards the distal end of the barrel.

[0036] One advantage of this is that it further prevents the syringe from being reused.

[0037] In some embodiments, the plunger stopper comprises a thread profile.

[0038] One advantage of this is that it allows an under pressure to be formed if necessary.

[0039] Further advantages will be apparent from the detailed description as well as the appended dependent claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] These and other aspects, features and advantages of which the invention is capable, will be apparent and elucidated from the following description of non-limiting embodiments of the present invention, reference being made to the accompanying drawings, in which:

[0041] FIG. 1*a* is a longitudinal section view of a plunger rod according to one example;

[0042] FIG. 1*b* is an isometric view of the same plunger rod;

[0043] FIG. 2 is a longitudinal section view of different plunger rods according to six examples;

[0044] FIG. 3 shows a longitudinal section view and an isometric view of a syringe according to one example;

[0045] FIG. 4 shows a longitudinal section view and an isometric view of a syringe according to one example;

[0046] FIG. 5 shows a longitudinal section view and an isometric view of a syringe according to one example;

[0047] FIG. 6 shows a longitudinal section view and an isometric view of a syringe according to one example;

[0048] FIG. 7*a* shows an isometric view of a threaded adapter according to one example;

[0049] FIG. 7*b* shows a longitudinal section view of a syringe according to one example;

[0050] FIG. 8*a* shows a top view of a packaging of a plunger rod and syringe according to one example; and

[0051] FIG. 8*b* shows an isometric view of a packaging of a plunger rod and syringe according to one example.

DESCRIPTION OF NON-LIMITING EMBODIMENTS

[0052] Embodiments of the present invention will be described in more detail below with reference to the accompanying drawings in order for those skilled in the art to be able to carry out the invention. The invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. The embodiments do not limit the invention, but the invention is only limited by the appended patent claims. Furthermore, the terminology used in the detailed description of the particular embodiments illustrated in the accompanying drawings is not intended

to be limiting of the invention.

[0053] Referring to FIGS. **1a-b**, one example of a plunger rod **104** according to the teachings herein is shown. The plunger rod **104** comprises an activation plate **107** at its proximal end for applying force thereto.

[0054] The plunger rod **104** further comprises a rod **114** that preferably extends at least the length of a barrel **102** of a syringe **100** that it is intended to be inserted into. The cross-section A-A of the rod **114** shown in FIG. **1a** illustrates four supporting ribs **115** extending radially from the center of the rod **114**. A width of the rod **114** at the cross section A-A is preferably equal to or within 10% greater than the inner diameter of the barrel **102** of the syringe **100** that the plunger rod **104** is intended to be inserted into; in order to ensure a tight fit in the barrel **102** which is used to guide the plunger rod **104**. The lateral ends of supporting ribs **115** are beveled, for these supporting ribs to yield to fit snugly with the inner surface of the barrel **102**. In this way, the position of the rod **114** in the barrel **102** may be maintained when no pushing or pulling of the rod **114** in distal or proximal direction, respectively, is performed

[0055] The plunger rod **104** further comprises a push plate **105** at its distal end **105**, which is used to push a plunger stopper **103**, as shown in FIGS. **3** and **4**, in the barrel **102** toward a distal end **110** of the barrel **102** after a force is applied to the proximal end **107** of the plunger rod **104**.

[0056] With reference to FIG. **2**, six examples of a plunger rod **104** according to the teachings herein are shown. Each example has a different non-limiting embodiment of a spring portion **106** at the distal end **105** of the plunger rod **104**.

[0057] Embodiment Nr1 uses press fitting to insert a distal end **105** of the plunger rod **104** with two opposing supporting ribs **115** having a diameter larger than the inner diameter of the barrel **102** of the syringe **100**. This achieves a radial deformation of the supporting ribs **115** which will radially apply a spring force to the barrel **102**, thereby achieving a spring portion **106**. By changing the size of portions of the supporting ribs **115**, the applied spring force may be optimized to be large enough to ensure the plunger rod **104** sits tight against the barrel **102** of the syringe **100** and small enough to ensure that the operation of the syringe **100**, meaning the transfer of force from the plunger rod **104** to the plunger stopper **103** leading to a distal displacement, is not disturbed.

[0058] Embodiment Nr2 uses a similar radial extension of two opposing supporting ribs **115** in order to achieve a spring portion **106** at the distal end **105** of the plunger rod **104**. However, in this case the spring portion **106** comprises an axial slit **116**. An advantage of this is that it may be easier to optimize the applied spring force according to the limits above. The axial slit **116** is contained within the ribs **115**, such that the slit **116** does not open up into the periphery of the rod **114**. In this way, there is no risk for the slit to interact with a proximal end **111** of the barrel **102** upon insertion of the plunger rod **104** into the barrel **102**. If the slit **116** opened up into the periphery of the rod **114**, there would be a risk that a protrusion formed laterally of the slit **116**, pointing distally, would be deformed during demoulding to further increase the risk for the slit **116** to interact with the proximal end **111** of the barrel upon insertion of the plunger rod **104**.

[0059] Embodiment Nr3 uses a similar concept as the previous embodiment of a radial extension combined with an axial slit opening up into the push plate **105** to achieve a spring portion **106**. In this case the axial slit is placed in the middle of the rod **114**, separating the supporting ribs **115** from each other. In this embodiment the width of the push plate **105** is somewhat larger than the inner diameter of the barrel **102** in relaxed state, such that the periphery of the push plate **105** will interact with the inner surface of the barrel **102**, in compressed state to maintain the plunger rod **104** therein also when no pushing or pulling is performed. An advantage of this is that it may be easier to optimize the applied spring force according to the limits above. Supporting ribs **115** are provided with interaction knobs **117**. The interaction knobs **117** will, during insertion of the plunger rod **104** into the barrel **102** interact with the inner surface of the barrel **102** in a retaining manner, whilst simultaneously allowing for distal and proximal movement of the plunger rod **104** upon pushing and pulling, respectively, of the plunger rod **104**.

[0060] Embodiment Nr4 is most similar to embodiment Nr2, except that the slit **116** has been extended radially in the proximal direction of the plunger rod **104**, forming an interaction spoke **118** extending substantially proximally and somewhat radially from the supporting ribs **115**. Hence, the slit **116** opens up into the periphery of the rod **114** in the proximal direction, such that the spokes **118** anyhow will yield centrally when inserting the plunger rod **104** into the barrel **102**, without an increased risk of having the slit **116** interacting with the proximal end **111** of the barrel **102**. An advantage of this is that it may be easier to optimize the applied spring force according to the limits above.

[0061] Embodiment Nr5 is most similar to embodiment Nr3, except that the supporting ribs **115** are reinforced with interaction discs **119** instead of interaction knobs **117**. The interaction discs **119** connect the ribs **115** and are dimensioned and configured to interact with the inner surface of the barrel **102** when the plunger rod **104** is displaced within the barrel **102**. The interaction discs **119** will, during insertion of the plunger rod **104** into the barrel **102** interact with the inner surface of the barrel **102** in a retaining manner, whilst simultaneously allowing for distal and proximal movement of the plunger rod **104** upon pushing and pulling, respectively, of the plunger rod **104**. An advantage of this is that it may be easier to optimize the applied spring force according to the limits above.

[0062] Embodiment Nr6 is most similar to embodiment Nr4, except that instead of a spoke **118** the ribs are provided with a set of reeds **120**. The reeds **120** are thinner than the spokes **118**, and will conform with the inner surface of the barrel **102**. The reeds **120** are directed substantially radially and somewhat proximally. An advantage of this is that it may be easier to optimize the applied spring force according to the limits above.

[0063] FIG. 3 illustrates one example of a syringe **100** comprising an example plunger rod **104** corresponding to embodiment Nr2 from FIG. 2. As the plunger rod **104** is inserted into a proximal end **111** of a barrel **102** axially along an axis **1000**, the spring portion **106** at the distal end **105** of the plunger rod **104** is displaced and a spring force is applied radially to the barrel **102**.

[0064] The syringe **100** comprises a barrel **102** with a needle **101** attached to a distal end **110** of the barrel **102**. The needle **101** is mechanically and hermetically attached to the syringe as known in the art, such as molded or glued, around the rear end of a hollow needle, whereby liquid passage is allowed in both directions, from the rear end of the needle unit to and through a needle tip of the hollow needle. The needle **101** is usually very thin, with a surface diameter of 0.3-0.6 mm. The barrel **102** of the syringe **100** may be made of any material suitable for containing a drug, but is preferably made of glass. The glass is preferably transparent. The barrel **102** preferably comprises a proximal end **111** which is adapted to act as a counter-balance to the force applied to the proximal end **107** of the plunger rod **104**. The barrel **102** is adapted to contain a drug. The drug may be in liquid form to be injected into a patient or user. The barrel **102** has a distal end **110**, to which the needle **101** is attached, and a tubular wall extending proximally from the distal end. The syringe **100** extends along an axis **1000** which defines the direction in which the barrel **102** extends, e.g. the barrel **102** extends along the axis **1000**.

[0065] A plunger stopper **103** is disposed within the barrel **102**. The plunger stopper **103** is adapted to be in fluid-tight engagement with the tubular wall of the barrel **102**. By being in fluid-tight engagement with the tubular wall of the barrel **102**, the drug contained within the barrel **102** is prevented from leaking out of the syringe **100** in any other direction than through the needle **101** at the distal end **110** of the barrel **102**. The plunger stopper **103** is disposed within the barrel **102** after the syringe **100** has been filled with the drug.

[0066] The plunger stopper **103** may however regardless of the thread profile **105b** be adapted to receive force along the axis **1000** from a plunger rod **104** without a thread profile **105a**. By displacing the plunger stopper **103** and the plunger rod **104** along the axis **1000** in a direction within the barrel **102** towards the distal end **110** with the attached needle **101**, injection of the drug is achieved. When the plunger stopper **103** and the plunger rod **104** have been moved as far as

possible towards the distal end **110** of the barrel **102**, almost all drug contained within the syringe **100** have been released and injected to the user. Since the syringe **100** is constructed such that dead space is minimized, almost all of the valuable drug will be released. After the drug has been injected, the plunger rod **104** may be discarded separate from the syringe **100**. This is advantageous as used syringes need to be disposed of in special containers, which need to be handled in a safe and costly manner. Therefore, by separating the plunger rod **104**, it will no longer take up unnecessary space in these containers and also be more efficiently recycled. In one embodiment according to the teachings herein, the syringe **100** contains the drug. The drug may need to be refrigerated, which means that being able to pack the syringes **100** efficiently is even more important. This is yet another advantage of being able to keep the plunger rod **104** separate.

[0067] FIG. 4 illustrates another example of a syringe **100** comprising an example plunger rod **104**. In this embodiment, the plunger stopper **103** has a hollow portion **105c**. As long as the proximal end of the plunger stopper **103** may receive the transmitted force of the plunger rod **104**, the plunger stopper **103** is within the scope of the amended claims. Thusly, the center of the plunger stopper **103** may be hollow to conserve materials or, as seen in FIG. 4, the proximal end of the plunger stopper **103** may only comprise an outer ridge.

[0068] FIG. 5 illustrates another example of a syringe **100** comprising an example plunger rod **104**. In this embodiment the plunger stopper **103** has a thread profile **105b** which is adapted to receive a piece with a thread profile **105a** that engages with the thread profile **105b** of the plunger stopper **103**. A thread is a helical structure used to convert between rotational and linear movement or force, and the thread is a ridge wrapped around a cylinder or cone in the form of a helix. The plunger stopper **103** according to the teachings herein is thus adapted to receive such a cone or cylinder with a thread profile **105a** configured to engage with the thread profile **105b** of the plunger stopper **103**.

[0069] FIG. 6 illustrates another example of a syringe **100** comprising an example plunger rod **104**, this time corresponding to embodiment Nr3 from FIG. 2.

[0070] FIGS. 7a-b illustrate one example of a syringe **100** comprising an example threaded adapter **121** comprising a thread profile **105a**. The thread profile **105a** of the threaded adapter **121** engages with the thread profile **105b** of the plunger stopper **103**. Because the plunger stopper **103** and the threaded adapter **121** have engaging thread profiles **105b**, **105a**, the two parts may be effectively attached to each other. The plunger stopper **103** represents an external thread and the threaded adapter **121** represents an internal thread. The engaging thread profiles **105b**, **105a** of the plunger stopper **103** and the threaded adapter **121** work as a screw with a matching bolt. By using a fastening element, such as engaging thread profiles **105a**, **105b**, the fastening element is relatively easy and cheap to produce, without the need to add any additional materials or components except the plunger stopper **103** and the threaded adapter **121**.

[0071] By engaging, and thereby attaching, the plunger stopper **103** and the threaded adapter **121** with each other by the engaging thread profiles **105b**, **105a**, the two parts may be displaceable together along the axis **1000** within said barrel **102**. By displacing the plunger stopper **103** and the threaded adapter **121** along the axis **1000** in a direction within the barrel **102** towards the distal end **110** with the attached needle **101**, injection of the drug is achieved. The thread engagement between the threaded adapter **121** and the plunger stopper **103** may allow for a displacement long the axis **1000** in a direction within the barrel **102** towards the proximal end **111** of the barrel **102**.

[0072] This may be used during manufacturing to form an under pressure which may be beneficial. It is also possible to use this threaded adapter **121** in order to make the syringe **100** reusable. The threaded adapter **121** should be integral in the assembly/filling machine.

[0073] If the need exists to create a negative pressure in the syringe **100** (aspirate), after the plunger stopper **103** has been fitted into the syringe **100**. Then it is possible to equip the mounting machine with a stainless steel threaded adapter **121**, which then thread in the internal thread **105b** of the plunger stopper **103**, which can cause a negative pressure, pulling the plunger stopper **103**

backwards (aspirate), which in turn creates the negative pressure in the area where the medication is located in the syringe **100**. This in turn should prevent leakage through the needle. The last step is to turn the thread rotation for the threaded adapter **121**, and thread it out of the plunger stopper **103**. This station is easy to integrate into the prefilled mounting machine.

[0074] FIGS. **8a-b** illustrate one example of a packaging of a plunger rod **104** and syringe **100** according to this disclosure. Because the plunger rod **104** is separate from the syringe **100**, they may be packed side-by-side in an unconventional manner, as shown. This conserves space and allows for more efficient storage and shipping of the product, which is specifically asked for with regard to pre-filled syringes, since they almost exclusively are stored and transported in cold storage. Another advantage of this packaging solution is that it is less susceptible to outside forces damaging the product or accidentally injecting a portion of the drug of a pre-filled syringe **100** into its packaging. A further advantage of this packaging solution is that the center of the package is closer to the center of the syringe **100** than conventional solutions, which means that the product may be more recognizable and helps a costumer or medical professional identify a specific brand or dosage.

[0075] One advantage of the described embodiments is that the costs for the manufacturing methods used to produce the syringes according to the teachings herein will not be increased, while still assuring that the syringes are not possible to reuse.

[0076] Further, the invention has mainly been described with reference to a few embodiments. However, as is readily understood by a person skilled in the art, other embodiments than the ones disclosed above are equally possible within the scope of the invention, as defined by the appended claims.

[0077] In the claims, the term “comprises/comprising” does not exclude the presence of other elements or steps. Furthermore, although individually listed, a plurality of means, elements or method steps may be implemented by e.g. a single unit or processor. Additionally, although individual features may be included in different claims, these may possibly advantageously be combined, and the inclusion in different claims does not imply that a combination of features is not feasible and/or advantageous. In addition, singular references do not exclude a plurality. The terms “a”, “an”, “first”, “second” etc. do not preclude a plurality. Reference signs in the claims are provided merely as a clarifying example and shall not be construed as limiting the scope of the claims in any way.

Claims

1. A syringe comprising: a barrel with a needle arranged at a distal end of the barrel, the barrel being adapted to contain a drug, the barrel having a tubular wall extending proximally from the distal end, whereby the barrel extends along an axis; a plunger stopper disposed within the barrel, the plunger stopper being adapted to be in fluid-tight engagement with the tubular wall of the barrel, the plunger stopper being displaceable along the axis within the barrel; wherein the barrel includes an opening at a proximal end of the barrel; a plunger rod with an elongated rod having a proximal end adapted to receive an applied force along the axis, and a distal end adapted to transmit the applied force along the axis; wherein the plunger rod is adapted to be inserted into the barrel extending along the axis, the plunger rod being adapted to transmit force along the axis to the plunger stopper; and a radial spring mechanism provided on the elongated rod, the radial spring mechanism adapted to interact with an inner surface of the barrel; wherein the radial spring mechanism is located radially outward from a center portion of the elongated rod, such that the center portion of the elongated rod extends through the radial spring mechanism; the plunger rod extending along the axis; wherein the plunger stopper is adapted to receive force along the axis from the plunger rod; and wherein the plunger rod is not connected to the plunger stopper, such that the plunger stopper cannot be pulled back by the plunger rod.

2. The syringe according to claim 1, wherein the plunger rod is made of a thermoplastic polymer having a Young's modulus within the interval 500 MPa to 2000 MPa.
 3. The syringe according to claim 1, wherein the elongated rod has a cross-shaped cross-section comprising four supporting ribs extending radially from a center of the elongated rod, and wherein at least one of the supporting ribs includes the radial spring mechanism.
 4. The syringe according to claim 3, wherein the four supporting ribs include interaction knobs.
 5. The syringe according to claim 3, wherein the four supporting ribs include interaction discs.
 6. The syringe according to claim 3, wherein the radial spring mechanism includes a spring portion on the distal end of the elongated rod that is adapted to be displaced radially from a relaxed position once the plunger rod is inserted into the barrel of the syringe.
 7. The syringe according to claim 6, wherein at least one of the four supporting ribs includes the spring portion.
 8. The syringe according to claim 7, wherein the spring portion includes an axial slit.
 9. The syringe according to claim 8, wherein the axial slit is arranged in a middle of the elongated rod.
 10. The syringe according to claim 8, wherein the axial slit is contained within the at least one supporting rib.
 11. The syringe according to claim 7, wherein the spring portion includes an interaction spoke.
 12. The syringe according to claim 7, wherein the spring portion includes a set of reeds.
 13. The syringe according to claim 1, wherein the syringe contains the drug.
 14. The syringe according to claim 1, wherein the plunger stopper is made of an elastomer having a Young's modulus within the interval 25 MPa to 300 MPa.
 15. The syringe according to claim 1, wherein the plunger stopper is adapted to only be displaceable along the axis towards the distal end of the barrel.
 16. The syringe according to claim 1, wherein the plunger stopper includes a thread profile.
 17. A syringe comprising: a barrel with a needle arranged at a distal end of the barrel, the barrel being adapted to contain a drug, the barrel having a tubular wall extending proximally from the distal end, whereby the barrel extends along an axis; a plunger stopper disposed within the barrel, the plunger stopper being adapted to be in fluid-tight engagement with the tubular wall of the barrel, the plunger stopper being displaceable along the axis within the barrel; wherein the barrel includes an opening at a proximal end of the barrel; a plunger rod with an elongated rod having a proximal end adapted to receive an applied force along the axis, and a distal end adapted to transmit the applied force along the axis; wherein the elongated rod has a plurality of supporting ribs extending radially from a center of the elongated rod; wherein the plunger rod is adapted to be inserted into the barrel, the plunger rod being adapted to transmit force along the axis to the plunger stopper; and a radial spring mechanism provided on the elongated rod, the radial spring mechanism adapted to interact with an inner surface of the barrel; the plunger rod extending along the axis; wherein the plunger stopper is adapted to receive force along the axis from the plunger rod; wherein the plunger rod is not connected to the plunger stopper, such that the plunger stopper cannot be pulled back by the plunger rod; and wherein the radial spring mechanism includes, on each of at least two of the supporting ribs, a spring portion adapted to be displaced radially from a relaxed position once the plunger rod is inserted into the barrel of the syringe and an axial slit.
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